

Customer Shopping Behaviour
CCDS 321 - Data Mining
Milestone 1

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Problem Statement

Nowadays, markets deal with a huge number of people daily, which might lead to inefficient decision-making and poor understanding of customer needs. Shopping Businesses face challenges in understanding customer behaviour, payment preferences, and purchase frequency. To handle this, the project aims to help businesses improve by including data cleaning, analysis, and pattern recognition that assist in predicting customers' behaviour whether they subscribe or not based on their Age, Gender, Shipping Type, and Item Purchased.

Objectives

- clean and preprocess the data before modeling.
- build a model that predicts whether the customer will subscribe or not.
- Identify the main factors that affect customer subscription status.
- evaluate the model's performance and interpret the results.

Hypotheses

- H1: Customers with higher purchase frequency are more likely to subscribe compared to customers with lower purchase frequency.
- H2: Customers with higher review ratings are more likely to subscribe than those with lower ratings.
- H3: Age and gender influence customer subscription behavior.

Project Questions

- Which factors most influence whether a customer subscribes or not?
- Can we predict if a customer will subscribe based on their shopping behavior?
- Are there clear differences between customers who subscribe and those who do not?
- Can customer segmentation (clustering) identify groups with a higher likelihood of subscription?

Data Source

We chose the Kaggle **Customer Shopping Behavior Dataset** as it provides a comprehensive set of well-structured variables that support analyzing customer shopping patterns and predicting whether a customer will subscribe or not.

The dataset includes several important features, such as:

- **Demographic Information:** Age, gender, and location of customers.
- **Shopping Attributes:** Purchased items, product category, size, color, and season.
- **Transaction Details:** Purchase amount (USD), shipping method, discount usage, and payment type.
- **Behavioral Indicators:** Subscription status, previous purchase history, purchase frequency, and customer review ratings.

The dataset can be accessed here: [Customer Shopping Behavior Dataset](#).